

CP260-2026 Final Project Specifications

Due Date: May 3 2026

The final project will involve Metric-Semantic reconstruction of a scene, given a set of posed images and point clouds. The reconstruction will then be used to answer questions related to spatial information of things in the scene.

Dataset:

The dataset is available for download from :
https://drive.google.com/file/d/1U8kTzhToFkHihi-6Qw0UTSO2M9_JLo3i/view?usp=sharing

It consists of a number of images of a desktop in files with names: frame_000xyz.png where xyz is the frame number.

There is another file named: poses.json – which has the poses of the images. Index is 'xyz'.

Your task:

Use these to create a representation of the scene to be able to handle the following down stream tasks:

a) ~~View synthesis from a novel view point.~~

~~Evaluation: The set of new view points will be given during evaluation.
Photometric reconstruction error will be used to score the quality of reconstructions.~~

b) Identify the pose of these parts of the desktop in the scene :

- a. Power socket
- b. Ethernet socket

Baseline Evaluation: The pose from your system will be checked against a ground truth.

Bonus Evaluation:

Pose for new parts will be asked for and checked against ground truth.

Coolness Evaluation: Any additional features/capabilities

Submission:

a) Github link to your project with

- a. /src subdirectory containing your source files
- b. README.md containing a detailed description of the project
- c. Any other folders
- d. /docs containing a pdf of the project report and any other documentation including instruction for running your code.

b) Project report should have these following components:

- a. Problem description
- b. Literature review of State of the Art
- c. Your system architecture and the rational for the design choices
- d. Implementation details – highlighting any interesting aspects.
- e. Experimental results
- f. Conclusions
- g. References
- h. Bonus: Any new ideas that were experimented with

Scoring:

50% of the course marks will be for this final project.

Within the project, marks will be apportioned as per the following

30% Report

20% Github repo with working code and proper documentation

15% Baseline Evaluation

25% Bonus Evaluation

10% Coolness Evaluation